AWS NETWORKING PRACTICAL

Build a Bastion Host on AWS



A step-by-step lab: VPC, subnets, routing, security groups, EC2 instances and SSH access to a private server.

Beginner-friendly • AWS Console + Windows PowerShell + Linux

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What you will build

By the end, you will securely reach a private EC2 instance through a Bastion Host.

1. Network foundation

Create one VPC, a public subnet, a private subnet, an Internet Gateway and route tables.

2. Access control

Use separate security groups to permit administrator-to-bastion and bastion-to-private SSH.

3. Secure connection

Launch two Ubuntu EC2 instances and connect from Windows 11 using SSH agent forwarding.

Lab approach: first build everything manually in the AWS Console, then verify connectivity with commands. Later, recreate the same design using Terraform.

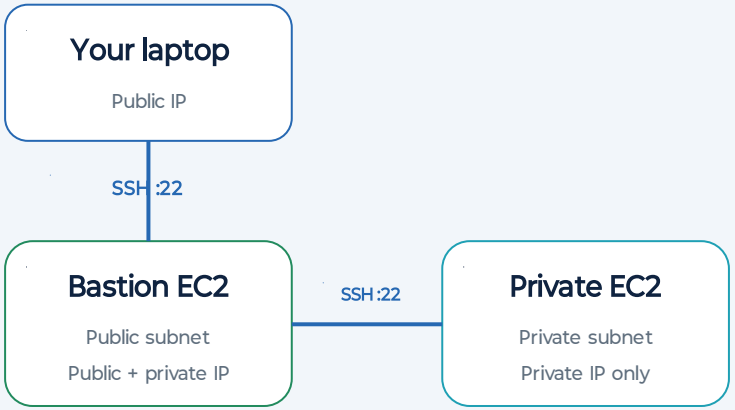
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1. What is a Bastion Host?

CONCEPT

A Bastion Host is a controlled gateway used to access servers in a private network.

A Bastion Host (also called a jump server) is an EC2 instance placed in a public subnet. Administrators connect to it first, then use it to reach private instances.



Why use it?

- Private servers need no public IP
- Central point for controlled administration
- Limit SSH access to trusted sources

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2. Lab architecture and IP plan

DESIGN

Use one VPC with two subnets in the same Availability Zone for this introductory lab.

Resource	Name	CIDR / IP
VPC	Bastion-VPC	10.0.0.0/16
Public subnet	Public-Subnet	10.0.1.0/24
Private subnet	Private-Subnet	10.0.2.0/24
Bastion EC2	Bastion-Server	10.0.1.x + public IP
Private EC2	Private-Server	10.0.2.x; no public IP

Region example

Mumbai: ap-south-1

Example AZ: ap-south-1a

Required services

Amazon VPC • EC2 • Internet Gateway

SSH client on Windows 11

CIDR reminder: /16 provides the VPC address range; /24 gives each subnet 256 IPv4 addresses (AWS reserves 5 addresses in each subnet).

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3. Create the VPC

AWS Console → VPC → Your VPCs → Create VPC → VPC only

- Name tag: Bastion-VPC
- IPv4 CIDR: 10.0.0.0/16
- IPv6 CIDR: No IPv6 CIDR block
- Tenancy: Default
- Choose Create VPC.

What is a VPC?

A logically isolated network in AWS. You choose its IP range and control subnets, routes and network access.

Check: the new VPC should appear in the VPC list with the CIDR 10.0.0.0/16.

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4. Create public and private subnets

VPC Console → Subnets → Create subnet

Field	Public subnet	Private subnet
VPC	Bastion-VPC	Bastion-VPC
Name	Public-Subnet	Private-Subnet
Availability Zone	ap-south-1a	ap-south-1a
IPv4 CIDR	10.0.1.0/24	10.0.2.0/24

Public subnet

Hosts the Bastion EC2 instance. It will have a route to the Internet Gateway.

Private subnet

Hosts the internal EC2 instance. It has no direct internet route in this basic lab.

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5. Create and attach an Internet Gateway

VPC Console → Internet Gateways → Create internet gateway

- Name it Bastion-IGW.
- Create the Internet Gateway.
- Select it → Actions → Attach to a VPC.
- Choose Bastion-VPC.
- Click Attach internet gateway.

What does an IGW do?

It provides a target for internet-bound routes and enables internet connectivity for eligible resources with public IPv4 addresses and suitable security rules.

An Internet Gateway alone does not make a subnet public. The subnet also needs a route to the IGW.

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6. Configure route tables

Routes decide where network traffic is sent.

Route table	Destination	Target	Subnet association
Public-RT	10.0.0.0/16	local	Public-Subnet
Public-RT	0.0.0.0/0	Bastion-IGW	Public-Subnet
Private-RT	10.0.0.0/16	local	Private-Subnet

Public-RT

Create route table in Bastion-VPC. Add default route 0.0.0.0/0 → Bastion-IGW. Associate Public-Subnet.

Private-RT

Create route table in Bastion-VPC. Keep the local route only. Associate Private-Subnet.

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7. Enable public IP assignment for the public subnet

VPC → Subnets → select Public-Subnet → Actions → Edit subnet settings

- Enable Auto-assign public IPv4 address on Public-Subnet.
- Save the subnet setting.
- Keep auto-assign public IPv4 disabled on Private-Subnet.
- When launching the Bastion EC2, verify that a public IPv4 address is assigned.

Important distinction

A public subnet is defined by its route to an Internet Gateway. A public IPv4 address is also needed for direct IPv4 internet reachability.

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8. Create the security groups

Security groups are stateful virtual firewalls attached to network interfaces.

Security group	Direction	Protocol / port	Source or destination
Bastion-SG	Inbound	TCP 22 (SSH)	Your public IP /32 only
Bastion-SG	Outbound	TCP 22 (SSH)	Private subnet 10.0.2.0/24 (optional restriction)
Private-Server-SG	Inbound	TCP 22 (SSH)	Bastion-SG security group ID
Private-Server-SG	Outbound	Default outbound	As required for lab

For the private instance inbound rule, choose Bastion-SG as the source security group. Do not type the group name as a CIDR. Security-group references are supported within the same VPC.

Avoid SSH from 0.0.0.0/0. Restrict administrator access to your current public IP /32.

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9. Create an SSH key pair

SECURITY

EC2 Console → Network & Security → Key Pairs → Create key pair

- Name: Bastion-Key
- Key pair type: RSA
- Private key format: .pem
- Create key pair and securely save the downloaded file.
- For this lab, the same key can be used for both instances.

Protect the private key

Never share the .pem file or commit it to GitHub. Keep it on your administrator workstation and protect access to it.

Bastion-Key.pem

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10. Launch the Bastion EC2 instance

EC2 → Instances → Launch instance

Setting	Value
Name	Bastion-Server
AMI	Ubuntu Server 24.04 LTS
Instance type	t3.micro (check eligibility and pricing)
Key pair	Bastion-Key
VPC / subnet	Bastion-VPC / Public-Subnet
Auto-assign public IP	Enable
Security group	Bastion-SG

After launch

Wait for Instance state = Running and both status checks to pass. Record its public IPv4 and private IPv4 addresses.

The Bastion Host should have:

- **Public IPv4 address**
- **Private address in 10.0.1.0/24**

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11. Launch the private EC2 instance

Launch a second Ubuntu EC2 instance in the private subnet.

Setting	Value
Name	Private-Server
AMI	Ubuntu Server 24.04 LTS
Instance type	t3.micro (check eligibility and pricing)
Key pair	Bastion-Key
VPC / subnet	Bastion-VPC / Private-Subnet
Auto-assign public IP	Disable
Security group	Private-Server-SG

Expected result

Private-Server has a private IPv4 address (10.0.2.x) and no public IPv4 address.

Do not add an inbound SSH rule from the internet. SSH should be allowed only from Bastion-SG.

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12. Connect from Windows 11 to Bastion

Use PowerShell and the built-in OpenSSH client.

Open PowerShell and move to the folder containing the downloaded key:

```
cd "$HOME\Downloads"
```

Connect using the Bastion instance's public IPv4 address:

```
ssh -i .\Bastion-Key.pem ubuntu@BASTION_PUBLIC_IP
```

Replace BASTION_PUBLIC_IP with the actual public IPv4 address. If prompted to trust the host, verify the host and type yes.

Successful login

You should see a shell prompt similar to: ubuntu@ip-10-0-1-10:~$

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13. Use SSH agent forwarding

Forward authentication from your Windows machine; do not copy the private key onto the Bastion Host.

In PowerShell opened as Administrator, enable and start the SSH Agent service:

```
Get-Service ssh-agent | Set-Service -StartupType Automatic
Start-Service ssh-agent
```

In a normal PowerShell window, add the key and verify it:

```
ssh-add "$HOME\Downloads\Bastion-Key.pem"
ssh-add -l
```

Connect to Bastion with agent forwarding enabled:

```
ssh -A -i "$HOME\Downloads\Bastion-Key.pem" ubuntu@BASTION_PUBLIC_IP
```


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14. Connect from Bastion to the private EC2

Once logged in to Bastion, use the private instance's private IPv4 address.

Run this command inside the Bastion Host:

```
ssh ubuntu@10.0.2.X
```

Replace 10.0.2.X with the actual private IPv4 address of Private-Server. If prompted to trust the host, type yes after confirming the destination.

Expected prompt

ubuntu@ip-10-0-2-XX:~$

What happened?

Your laptop authenticated through the SSH agent; the Bastion connected to the private instance over the VPC local route.

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15. Verify connectivity

VALIDATION

Run these commands on the relevant Linux instance.

On the Bastion Host

hostname
hostname -I
ip route

On the Private Server

hostname
hostname -I
ip route

From Bastion, test SSH reachability to the private instance:

```
nc -zv 10.0.2.X 22
```

Ping may fail if ICMP is blocked. A failed ping alone does not prove SSH is broken; use the TCP port test and SSH itself.

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16. Troubleshooting common errors

Check the network path from the outside in.

Error / symptom	What to check
Connection timed out to Bastion	Public IPv4, Public-RT default route to IGW, IGW attachment, Bastion-SG inbound TCP 22 from your IP.
Permission denied (publickey)	Correct username (ubuntu), correct key pair, correct local key and key permissions.
Private SSH times out	Private-Server-SG inbound TCP 22 source = Bastion-SG; Bastion outbound rules; private IP; local route.
Connection refused	SSH service may be stopped or not installed on the destination.
Agent contains no identities	Run ssh-add with the correct .pem path; verify with ssh-add -l.
No route to host	Check VPC, subnet association, IP address and routing.

On Ubuntu, check SSH service: sudo systemctl status ssh

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17. Useful Linux checks and key safety

Only run installation commands if the SSH server is missing.

```
sudo systemctl status ssh

sudo apt update
sudo apt install openssh-server -y
```

Private key handling

Keep the .pem file on your trusted workstation. Prefer agent forwarding or a managed access service over copying private keys to servers.

Security hardening ideas

Use a fixed administrator IP /32, least-privilege security groups, strong key protection, logging and patching. Consider a managed access option such as AWS Systems Manager Session Manager.

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18. Interview questions – quick answers

INTERVIEW PREP

Review the core concepts after completing the lab.

Question	Short answer
What is a Bastion Host?	A controlled gateway used to access private network resources.
Where is it deployed?	Usually in a public subnet with a route to an Internet Gateway.
Which port is used for Linux SSH?	TCP port 22 by default.
Does the private EC2 need a public IP?	No; it is reached using its private IP through the VPC.
What is a jump server?	Another term for a Bastion Host.
What is SSH agent forwarding?	It lets a remote host use the local SSH agent for authentication without copying the private key.
Is NAT Gateway required for this lab?	No, not for Bastion-to-private SSH within the same VPC.
What is a modern alternative?	AWS Systems Manager Session Manager.

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19. Bastion Host vs Systems Manager Session Manager

BEST PRACTICE

For production, evaluate whether you need a traditional internet-facing jump host.

Area	Traditional Bastion	Systems Manager Session Manager
Access path	SSH to public gateway, then private host	Managed session through AWS Systems Manager
Inbound SSH	Usually TCP 22 to Bastion	Can work without inbound SSH
Public IP	Bastion typically needs public reachability	Managed connectivity can work without public IP when prerequisites are met
Authentication	SSH key / SSH agent	IAM permissions and supported session mechanisms
Operations	You patch and maintain Bastion	AWS manages the service; you manage instance prerequisites and IAM

Session Manager prerequisites still matter: the instance needs a supported SSM Agent, IAM permissions and network connectivity to Systems Manager endpoints (directly or through configured endpoints).

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20. Clean up resources after the lab

Terminate resources when you finish to avoid ongoing AWS charges.

- Terminate Bastion-Server and Private-Server EC2 instances.
- Detach and delete Bastion-IGW.
- Delete the two subnets.
- Delete the custom route tables.
- Delete the security groups after their network interfaces are removed.
- Delete Bastion-VPC.
- Delete the key pair if it is no longer needed.
- Review AWS Billing and Cost Explorer for remaining resources.

Cost reminder

EC2, public IPv4 addresses, NAT Gateways and other resources may incur charges. Check current AWS pricing and Free Tier eligibility before launching.

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21. Practical completion checklist

Tick each item as you complete the lab.

- Create Bastion-VPC
- Create Public-Subnet and Private-Subnet
- Attach Internet Gateway and configure routes
- Configure subnet associations and public IP setting
- Create Bastion-SG and Private-Server-SG
- Create Bastion-Key
- Launch Bastion EC2 and Private EC2
- SSH from Windows to Bastion
- SSH from Bastion to private EC2
- Verify connectivity and clean up

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Lab complete: what you learned

You built a VPC with public and private subnets, configured routes and security groups, launched two EC2 instances, and securely connected to a private Linux server through a Bastion Host.

Next step

Rebuild this architecture using Terraform or CloudFormation.

Production direction

Explore Systems Manager Session Manager and centralized logging.

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